Ripasso Analisi II

Sia $f(x,y,z)$, $\quad df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$

Sia $\vec{r}(x,y,z)$, $\quad d\vec{r} = dx\,\vec{u}_x + dy\,\vec{u}_y + dz\,\vec{u}_z$

$\vec{\nabla} f = \frac{\partial f}{\partial x}\vec{u}_x + \frac{\partial f}{\partial y}\vec{u}_y + \frac{\partial f}{\partial z}\vec{u}_z$

$\Rightarrow df = \vec{\nabla} f \cdot d\vec{r} = |\vec{\nabla} f|\, dr \cos\theta$

- Il gradiente è perpendicolare alle superfici di livello ($f = cost$)

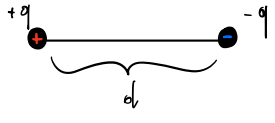
$\Rightarrow \frac{df}{dr} = |\vec{\nabla} f| \cos\theta$ derivata direzionale

$dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$

$\vec{E}\cdot d\vec{l} = E_x dx + E_y dy + E_z dz$

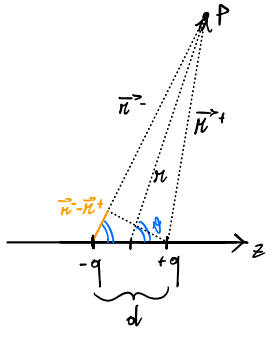
$\Rightarrow \begin{cases} E_x = \frac{\partial V}{\partial x} \\ E_y = \frac{\partial V}{\partial y} \\ E_z = \frac{\partial V}{\partial z} \end{cases} \Rightarrow \boxed{\vec{E} = -\vec{\nabla} V}$

DIPOLO ELETTRICO



$q_{tot} = +q - q = 0$

Qual'è il potenziale del dipolo?



$V(r) = V^+(r) + V^-(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^+} + \frac{1}{4\pi\varepsilon_0}\frac{-q}{r^-}$

$= \frac{q}{4\pi\varepsilon_0}\cdot\frac{r^- - r^+}{r^- \cdot r^+} \overset{(\approx)}{\underset{r \gg d}{\downarrow}} \frac{q}{4\pi\varepsilon_0}\frac{d\cos\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\vec{r}}{r^3}$

$\vec{p} = q\, d\, \vec{u}_z$ Momento di dipolo

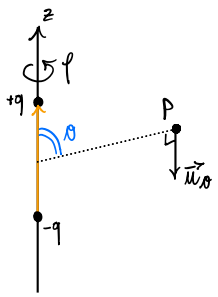


$$\Rightarrow \boxed{V(r) = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\vec{p}\cdot\vec{r}}{r^3}}$$ Potenziale rispetto al dipolo

oss. $V(r) \propto \frac{1}{r^2}$ $\Rightarrow$ decade molto + velocemente di quello della carica puntiforme $\left(\propto \frac{1}{r}\right)$

Voglio calcolarne il gradiente:

$$df = \frac{\partial f}{\partial r} dr + \frac{\partial f}{\partial \theta} d\theta + \frac{\partial f}{\partial \varphi} d\varphi$$



$$d\vec{r} = dr\,\vec{u}_r + r\,d\theta\,\vec{u}_\theta + r\sin\theta\, d\varphi\,\vec{u}_\varphi$$

$$df = \vec{\nabla} f \cdot d\vec{r}$$

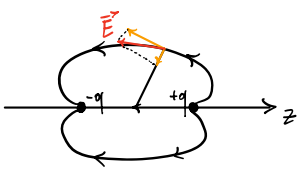
$$\Rightarrow \vec{\nabla} f = \frac{\partial f}{\partial r}\vec{u}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\vec{u}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \varphi}\vec{u}_\varphi$$

$$\Rightarrow \begin{cases} E_r = -\frac{\partial V}{\partial r} = +\frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{r^3} \\ E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta} = +\frac{1}{r}\frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{r^3} \\ E_\varphi = -\frac{1}{r\sin\theta}\frac{\partial V}{\partial \varphi} = 0 \Rightarrow \text{poiché } \varphi \text{ non compare in } V(r) \end{cases}$$

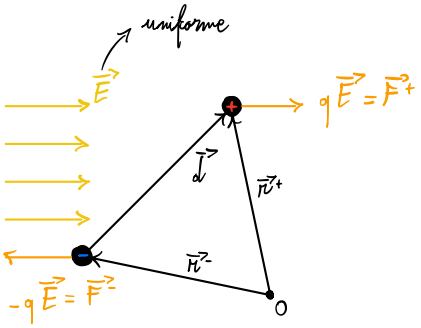
$$\boxed{\vec{E} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2p\cos\theta}{r^3}\vec{u}_r + \frac{1}{4\pi\varepsilon_0}\cdot\frac{p\sin\theta}{r^3}\vec{u}_\theta}$$ Campo elettrostatico del dipolo

$\Rightarrow \vec{E} \propto \frac{1}{r^3}$ $\Rightarrow$ decade + velocemente di quella della carica puntiforme

- per $\theta = 0 \Rightarrow \vec{E} = E_r\,\vec{u}_r$
- per $\theta = \frac{\pi}{2} \Rightarrow \vec{E} = E_\theta\,\vec{u}_\theta$



AZIONI MECCANICHE



$$\vec{F} = F^+ + F^- = 0 \implies \vec{a}_{CM} = 0$$

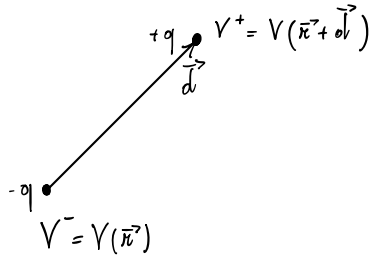
$$\vec{M}(O) = \vec{r}^+ \times \vec{F}^+ + \vec{r}^- \times \vec{F}^- = (\vec{r}^+ - \vec{r}^-) \times \vec{F}^+ = \vec{d} \times \vec{F}^+$$

$$= \vec{d} \times q\vec{E} = \vec{p} \times \vec{E}$$

$$\implies \boxed{\vec{M}(O) = \vec{p} \times \vec{E}}$$ Momento esterno (per il dipolo)

oss. Il momento esterno tende ad orientare il dipolo

• Consideriamo



$$E_p = E_p^+ + E_p^- = qV^+ - qV^- = q(V^+ - V^-) =$$

$$= q\left(V(\vec{r} + \vec{d}) - V(\vec{r})\right) \ominus q\,dV = q\,\vec{\nabla}V \cdot \vec{d} = \vec{\nabla}V \cdot \vec{p} = -\vec{p} \cdot \vec{E}$$

$\vec{d} \simeq d\vec{r}$

$$\implies \boxed{E_p = -\vec{p} \cdot \vec{E}}$$ Energia potenziale del dipolo

oss. Tutto ciò vale per campo uniforme!

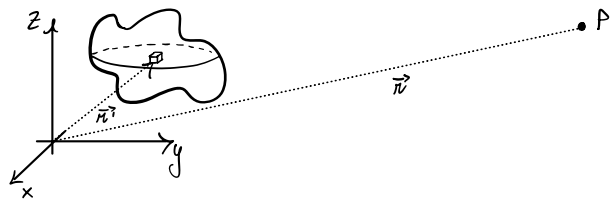
Nel caso $\vec{E} \neq cost$

$$\vec{F}_{ext} = -\vec{\nabla}E_p = \vec{\nabla}(\vec{p}\vec{E}) = (\vec{p} \cdot \vec{\nabla})\vec{E}$$

Espansione in Multipoli

$$V(r) = \frac{1}{4\pi\varepsilon_0} \int_\tau \frac{\rho(r)\, d\tau}{|\vec{r} - \vec{r}'|}$$

Se osservo da molto lontano?



$\vec{r}' \longrightarrow 0$ per un'osservatore molto distante

$$f(x', y', z') \simeq f(0) + \frac{\partial f}{\partial x'}\Big|_0 x' + \frac{\partial f}{\partial y'}\Big|_0 y' + \frac{\partial f}{\partial z'}\Big|_0 z' \qquad \text{(la considero come la sua espansione)}$$

$$f(x', y', z') = \frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}}$$

- $f(0) = \dfrac{1}{\sqrt{x^2+y^2+z^2}} = \dfrac{1}{r}$

- $\dfrac{\partial f}{\partial x'} = \dfrac{1}{2} \dfrac{2(x-x')}{\left[(x-x')^2 + (y-y')^2 + (z-z')^2\right]^{\frac{3}{2}}} = \dfrac{(x-x')}{|\vec{r} - \vec{r}'|^3}$

$$\Rightarrow \frac{\partial f}{\partial x'}\Big|_0 = \frac{x}{r^3}$$

$$\Rightarrow \frac{1}{|\vec{r} - \vec{r}'|} \simeq \frac{1}{r} + \frac{x x'}{r^3} + \frac{y y'}{r^3} + \frac{z z'}{r^3} = \frac{1}{r} + \frac{\vec{r} \cdot \vec{r}'}{r^3}$$

$$V(r) = \frac{1}{4\pi\varepsilon_0} \int_\tau \left\{ \frac{\rho(\vec{r}')}{r} + \frac{\rho(\vec{r}')\, \vec{r} \cdot \vec{r}'}{r^3} \right\} d\tau$$

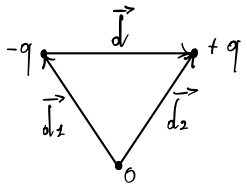
$$q = \int_\tau \rho(\vec{r}')\, d\tau$$

$$\vec{p} = \int_\tau \rho(\vec{r}')\, \vec{r}'\, d\tau \qquad \text{Perché?} \ *$$

$$\Rightarrow V(r) = \frac{1}{4\pi\varepsilon_0} \int_\tau \left[\frac{\rho(\vec{r}')}{r} + \frac{\rho(\vec{r}')\,\vec{r}\cdot\vec{r}'}{r^3} \right] d\tau = \underbrace{\frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r}\right.}_{\text{Monopolo}} + \underbrace{\left.\frac{\vec{p}\cdot\vec{r}}{r^3}\right]}_{\text{Dipolo}}$$

<u>OSS.</u> Il momento di dipolo è molto meno rilevante ($\propto \frac{1}{r^2}$). Diventa importante per $q = 0$

$\star$ $\vec{p} = \sum_i q_i \vec{d}_i$



$$\vec{p} = \vec{d}_1(-q) + \vec{d}_2 q = (\vec{d}_2 - \vec{d}_1) q = \vec{d}\, q$$

$\Rightarrow$ lo stesso vale per l'integrale